

Fig. 1 shows how the velocity v of a car varies with time t as the car approaches a road junction.

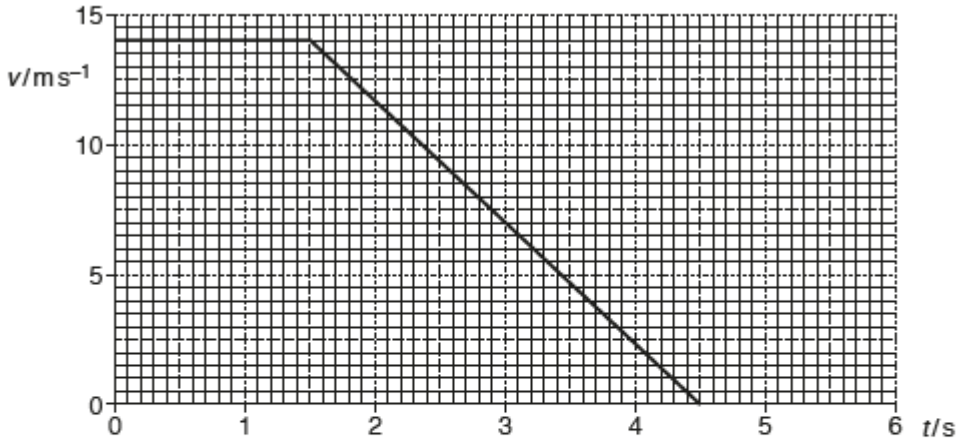


Fig. 1

- (a) Explain what feature of the graph shows the deceleration of the car and that the deceleration is constant after 1.5 s.

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.....
..... [2]

- (b) The driver of the car applies the brakes at a distance of 20 m from the 'stop line' at the junction.
Calculate the distance s of the car relative to the stop line when the car comes to a stop.

$s = \dots\dots\dots \text{m}$ [3]

Question		Answer	Marks	Guidance
1	(a)	Deceleration is indicated by the <u>negative gradient/slope</u> (between $t = 1.5$ to $t = 4.5$)	B1	
		A straight line/constant (negative) gradient (between $t = 1.5$ to $t = 4.5$) (indicates constant deceleration)	B1	Constant negative gradient scores two marks
	(b)	area under graph = displacement or distance	C1	ALLOW $s = \frac{(u+v)}{2}t$ or $s = ut + \frac{1}{2}at^2$ and $a = 14/3$
		$\frac{1}{2} \times 3 \times 14 = 21$ (m)	M1	$\frac{(14+0)}{2} \times 3$ OR 7×3
		1 (m)	A1	Allow ECF for mis read of t or v Do not accept $t = 4.5$ Ignore "-" sign
Total			5	